

Chapter 1: Computer Networks and the Internet

Exercise 1

This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

a) **The propagation delay:** $d_{\text{prop}} = d/s$:

M : meters as the distance between the two hosts

S : meters per second as the propagation speed along the link.

The propagation delay, d_{prop} , is the time it takes for a signal to travel from the sender to the receiver at the physical layer of the network. It is determined by the distance between the two hosts and the speed of signal propagation along the link.

b) $d_{\text{trans}} = L/R$

L : bits as the size of the packet to be transmitted

R : bits per second as the transmission rate of the link

The transmission time, often referred to as the transmission delay (d_{trans}), is the amount of time required to push all the packet's bits into the link. This delay depends on the size of the packet and the rate at which the link can transmit data.

c) The end-to-end delay = $d_{\text{prop}} + d_{\text{trans}} = d/s + L/R$

d) At time $t = d_{\text{trans}}$, the last bit of the packet is just leaving host A

e) If d_{prop} is greater than d_{trans} , at time $t = d_{\text{trans}}$, the first bit of the packet has not reached host B yet. It is still in transit.

f) If d_{prop} is less than d_{trans} , at time $t = d_{\text{trans}}$, the first bit of the packet has already reached host B.

g) $d/s = L/R$

$d = (L * S/R)$

$= (120 * 2.5 * 10^8) / 56000$

$= 535714.28571 \text{ meters}$

Exercise 2

In this problem, we consider sending real-time voice from Host A to Host B over a packet-switched network (VoIP). Host A converts analog voice to a digital 64 kbps bit stream on the fly. Host A then groups the bits into 56-byte packets. There is one link between Hosts A and B; its transmission rate is 2 Mbps and its propagation delay is 10 msec. As soon as Host A gathers a packet, it sends it to Host B. As soon as Host B receives an entire packet, it converts the packet's bits to an analog signal. How much time elapses from the time a bit is created (from the original analog signal at Host A) until the bit is decoded (as part of the analog signal at Host B)?

Given: Transmission rate: 2 Mbps, Propagation delay: 10 ms

Bitstream rate: 64kbps, Packet size: 56-byte* 8 bits/byte = 448 bits

Time elapsed = Packetization Delay + d_{trans} + d_{prop}

$$= 448/64000 + \text{Packet size in bits} / \text{Transmission rate} + 10 \text{ ms}$$

$$= 7\text{ms} + 448 \text{ bits} / 2\text{Mbps} + 10\text{ms}$$

$$= 7\text{ms} + 0.224\text{ms} + 10 \text{ ms}$$

$$= 17.224\text{ms}$$

Thus, It elapses from the time a bit is created from the original analog signal at host A until the bit is decoded as part of the analog signal at Host B, which is 17.224 ms

Exercise 3

Suppose users share a 3 Mbps link. Also suppose each user requires 150 kbps when transmitting, but each user transmits only 10 percent of the time. (See the discussion of packet switching versus circuit switching in Section 1.3)

a) Number of users = Total bandwidth/ Bandwidth per user = 3000kbps/150kbps = 20 users

b) Given that each user transmits only 10% of the time: $p = 0.1$

c) $P(X = n) = C(N, n)P^n(1-p)^{N-n} = C(20, n)(0.1)^n(0.9)^{20-n} = 0.10750 = 10.75\%$

d) $P(X \geq 21) = \sum_{n=21}^{120} \binom{120}{n} p^n (1-p)^{120-n} = 0.0079 = 0.79\%$

Exercise 4

Given parameters: $L = 1500\text{bytes} = 1500 * 8 \text{ bits} = 12,000 \text{ bits}$

Propagation Speed = $2.5 * 10^8 \text{ m/s}$ for all links

Transmission rate, $R_i = 2 \text{ Mbps} = 2 * 10^6 \text{ bps}$

$d_{proc} = 3 \text{ msec} = 0.003 \text{ seconds}$

Link Lengths: $d_1 = 5000\text{km} = 5000000\text{m}$, $d_2 = 4000,000\text{m}$, $d_3 = 1000,000\text{m}$

$$d_{trans,1} = d_{trans,2} = d_{trans,3} = 12000 \text{ bits} / (2 \times 10^6 \text{ m/s}) = 0.006 \text{ seconds}$$

$$d_{prop,1} = 5000000 \text{ m} / (2.5 \times 10^8 \text{ m/s}) = 0.02 \text{ seconds}$$

$$d_{prop,2} = 4000000 \text{ m} / (2.5 \times 10^8 \text{ m/s}) = 0.016 \text{ seconds}$$

$$d_{prop,3} = 1000000 \text{ m} / (2.5 \times 10^8 \text{ m/s}) = 0.004 \text{ seconds}$$

$$\begin{aligned} \text{Total end-to-end delay} &= \text{sum of } d_{trans,1,2,3} + d_{prop,1} + d_{prop,2} + d_{prop,3} + 2 \cdot d_{proc} \\ &= 3 \cdot 0.006 + 0.02 + 0.016 + 0.004 + 0.003 \cdot 2 \\ &= 0.064 \text{ seconds} \end{aligned}$$

Exercise 5

In the problem given we calculated: $d_{trans} = 0.006$, $d_{prop,1} = 0.02$, $d_{prop,2} = 0.016$, $d_{prop,3} = 0.004$

since the bits do not need to wait at switches:

$$\begin{aligned} D_{end-to-end} &= d_{trans} + d_{prop,1} + d_{prop,2} + d_{prop,3} \\ &= 0.006 + 0.02 + 0.016 + 0.004 \\ &= 0.046 \text{ seconds} \end{aligned}$$

Exercise 6:

Given parameters: Packet size: 1,500 bytes = 12,000 bits,

Link rate 2000000 bits per seconds

Time to finish = Remaining bits / Link rate = 6000 bits / 2000000 bits/sec = 0.003 seconds

Total time for 4 packets = 4 packets * (12000 bits / 2000000 bit/sec) = 4 * 0.006 seconds = 0.024 seconds

Total queuing delay = 0.003 seconds + 0.024 seconds = 0.027 seconds

General Formula for queuing delay: $Q = (N \cdot L - X) / R$,

where N: Number of packets in the queue,

L: length of a packet

$N \cdot L$: gives the total bits for all N packets, assuming N includes the currently transmitting packet.

X: Number of bits already transmitted in the currently being transmitted packet, so subtract X to account for what has already been transmitted in the current packet.

Exercise 7:

a) The transmission time for each packet is L/R . Therefore, the total queuing delay for all N packets is the sum of the transmission times for the first packet through the (N-1)th packet.

This can be calculated: Sum of queuing delays = $0 \cdot L/R + 1 \cdot L/R + 2 \cdot L/R + \dots + (N-1) \cdot L/R$

The sum of such a series is given by the formula: Sum = $N/2 \cdot (\text{first term} + \text{last term})$

Substituting the values:

$$\text{Sum of queuing delays} = N/2 \cdot ((N-1) \cdot L/R)$$

The average queuing delay is the total queuing delay divided by the number of packets, N:

$$\begin{aligned} \text{Average queuing delay} &= (N/2 \cdot ((N-1) \cdot L/R)) / N \\ &= (N-1)/2 \cdot (L/R) \end{aligned}$$

b) Now suppose that N such packets arrive at the link every LN/R seconds. This means that as soon as the last packet of a batch is transmitted, a new batch of N packets arrives. Therefore, every packet except the first one in each batch will experience a queuing delay.

The average queuing delay for a packet is still given by the formula from part (a), because the pattern of queuing delays within each batch is the same:

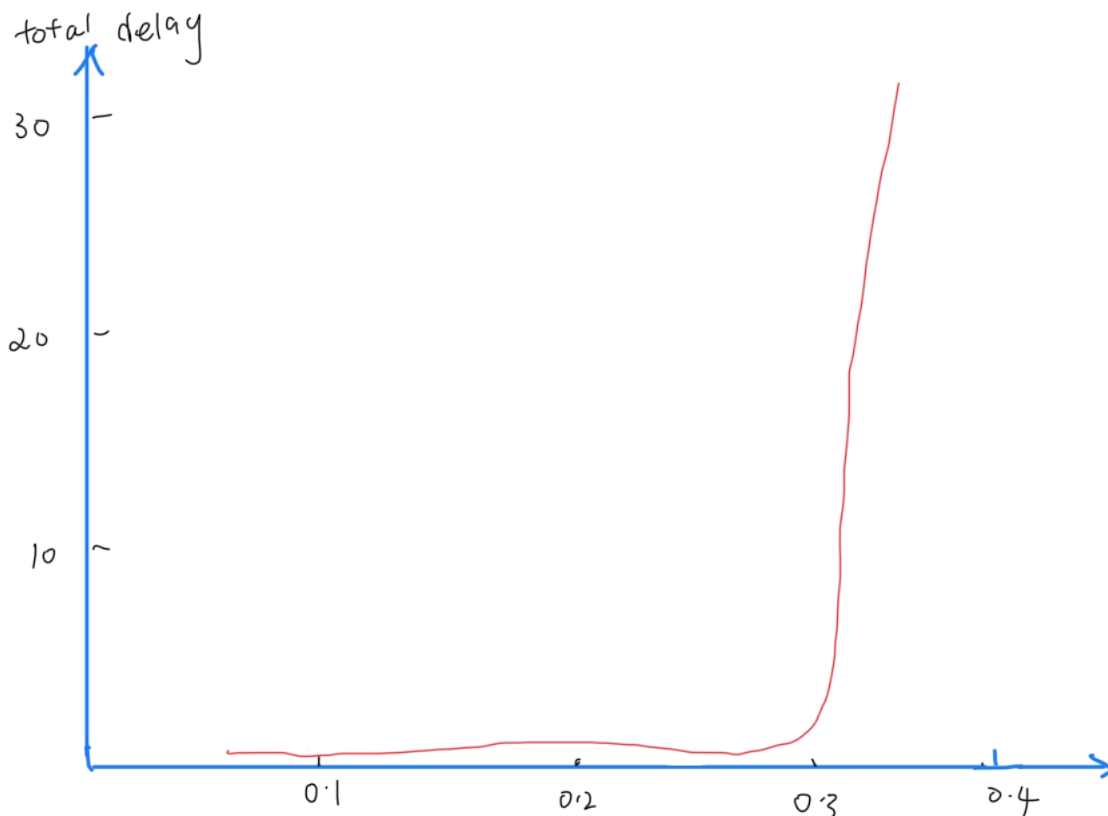
$$\text{Average queuing delay} = (N-1)/2 \cdot (L/R)$$

Exercise 8:

a) Since all packets consist of L bits, thus the transmission delay is L/R .

The total delay is queuing delay + transmission delay = $IL/R(1-I) + L/R = L / (R(1-I))$

b) Let L/R be x, thus the total delay is $x / 1-x$, let a be 3:



Exercise 9:

From problem 8 we know $D_{\text{queue}} = L / R(1-\rho)$, L is implicit as the size of one packet, and $R = L * \mu$, translating capacity in bits per second to packets per second thus:

$$D_{\text{queue}} = L * 1 / \mu(1-\rho) = \alpha / \mu / \mu(1 - \alpha / \mu) = \alpha / \mu^2 - \alpha$$

$$\text{Total Delay: } D_{\text{total}} = D_{\text{trans}} + D_{\text{queue}} = 1 / \mu + \alpha / \mu^2 - \alpha$$

Exercise 10:

Given parameters: Distance between A to B = 20,000,000 meters, $R = 2 \text{ Mbps}$

- $D_{\text{prop}} = 20,000,000 \text{ meters} / 2.5 * 10^8 = 0.08 \text{ s}$, Therefore $R * D_{\text{prop}} = 2 * 10^8 * 0.08 = 16,777,216 \text{ bits}$
- The bandwidth-delay product is the maximum number of bits that can be in the link and 16,777,216 bits which is smaller than 8,000,000 bits, thus the maximum number of bits in the link at any given time is 16,777,216 bits/
- The bandwidth-delay product is the maximum number of bits that can be in the link at a given time
- The width of a bit in the link = $20,000,000 / 16,777,216 \text{ bits} = 119.209 \text{ m}$, thus It is longer than a football field if the football field is less than 100 meters,
- General form:

$$\text{Width of a bit is } d / R * d_{\text{prop}} = d / (R * d/s) = s/R$$

Exercise 11:

- Propagation delay = distance/ propagation speed
 $= 36,000,000 / 2.4 * 10^8$
 $= 0.15 \text{ seconds}$
- Bandwidth-delay product($R * d_{\text{prop}}$): $R * d_{\text{prop}} = 10 \text{ Mbps} * 0.15 \text{ seconds}$
 $= 1,500,000 \text{ bits}$
- Time between photos: 1 minute = 60 seconds
 Link capacity: $R = 10 \text{ Mbps} = 10 * 10^6 \text{ bits/sec}$
 The minimum photo size: $x = R * \text{time between photos}$
 $= 10 * 10^6 \text{ bits/sec} * 60 \text{ seconds}$
 $= 600 * 10^6 \text{ bits}$
 $= 600 \text{ Megabits}$